Does bijectivity of g decide the Composite gain?

A short report on one experiment

The question in four sentences

Our benchmarks show Composite beating Direct by very different amounts, and we want to know what separates the cases. The proposition says that if g is *bijective*, observing the intermediates h tells you nothing you did not already learn from the objective y. So a natural guess is: **Composite helps when g is not bijective, and not otherwise.** This note tests that guess, and the answer is **no — bijectivity is not what decides it**.

1 What we built

A family of outer maps that are all the same shape of thing — products of linear factors — so that we can turn one property at a time:

$$g_m(z) = \prod_{i=1}^{m} (z - a_i), \qquad m = 1, \ldots, 5,$$

with the roots a_i spread across the range that h actually visits. Against each of these we put a map of the *same polynomial degree* that is bijective: z^3 and z^5 are strictly increasing, so they are just as nonlinear and just as steep, but nothing is lost. Plus e^z and $\sinh z$ as two more bijective references.

That matched pairing is the point. A degree-5 polynomial is both more non-bijective *and* a wilder function; if we only varied m we could never say which mattered.

That reasoning turns out to be unsound — the pairing does not hold model difficulty fixed after all, for a reason that rules out every one-dimensional design. Section efsec:inseparable works through it.

2 Measuring how non-bijective each one is

Rather than assume it from the degree, we count it. For each map, over the values of y the problem actually produces,

$$\bar{n} = \mathbb{E}\left|g^{-1}(y)\right| = \text{average number of } h \text{ values consistent with one observation.}$$

$\bar{n} = 1$ means bijective where it matters. The five bijective maps all come out at exactly 1.00; the others at 2.00, 2.82, 3.75 and 4.50. So the knob works, and it is measured rather than asserted.

3 Why we stopped measuring optimisation

We first compared the two models by how well they optimise. That turned out to be the wrong instrument, for a reason that cannot be fixed by choosing better roots:

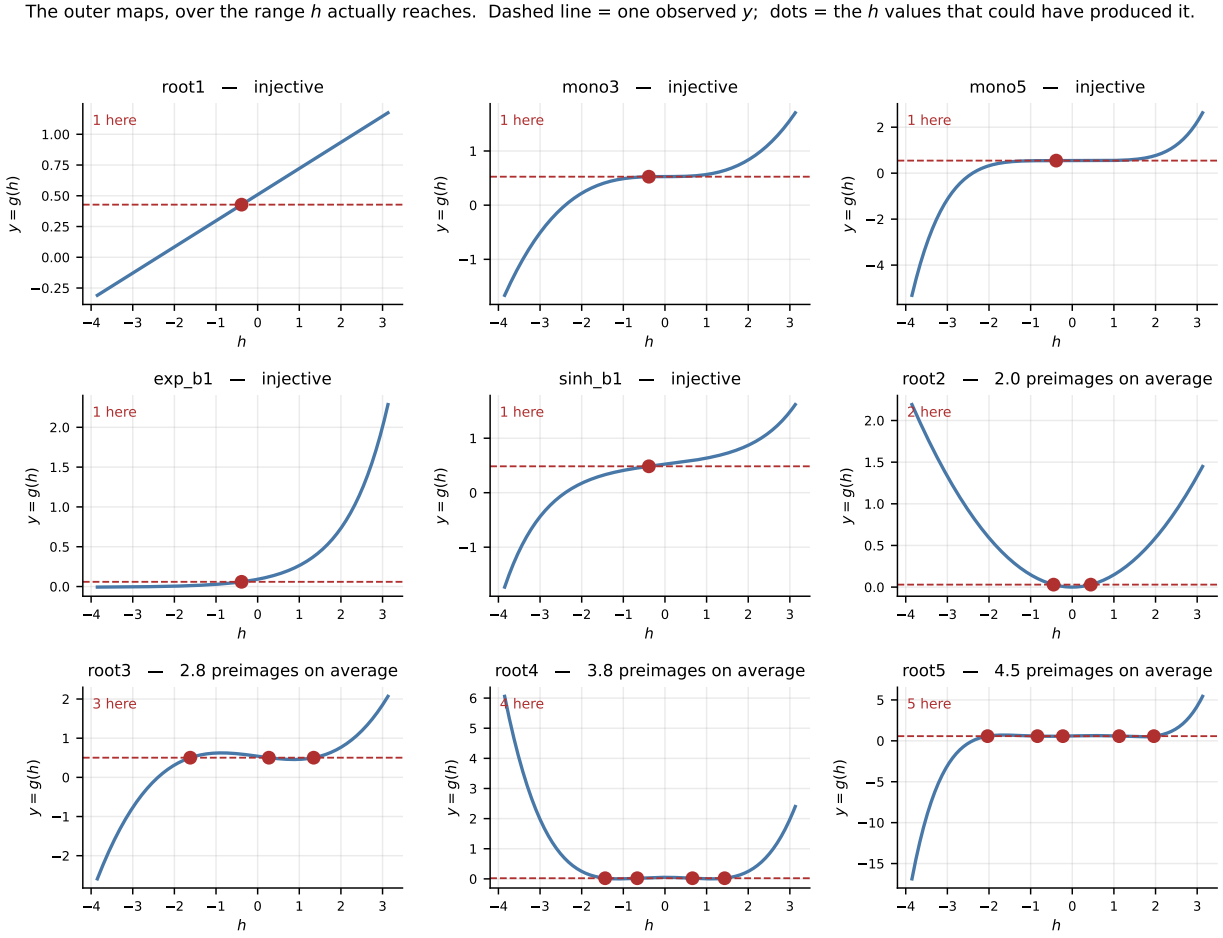


Figure 1: Every map in the family. The dashed line is one observed value of y; the dots are the values of h that could have produced it. One dot means y tells you h exactly — that is what bijective means. Several dots mean it does not.

A bijective g is monotone. A monotone function attains its minimum at an endpoint.

So a bijective map always puts the optimum at the edge of the reachable range, and a non-bijective one can put it in the middle. Bijectivity decides *what the optimisation problem is*, which means any comparison of optimisation performance between the two classes is confounded before it starts.

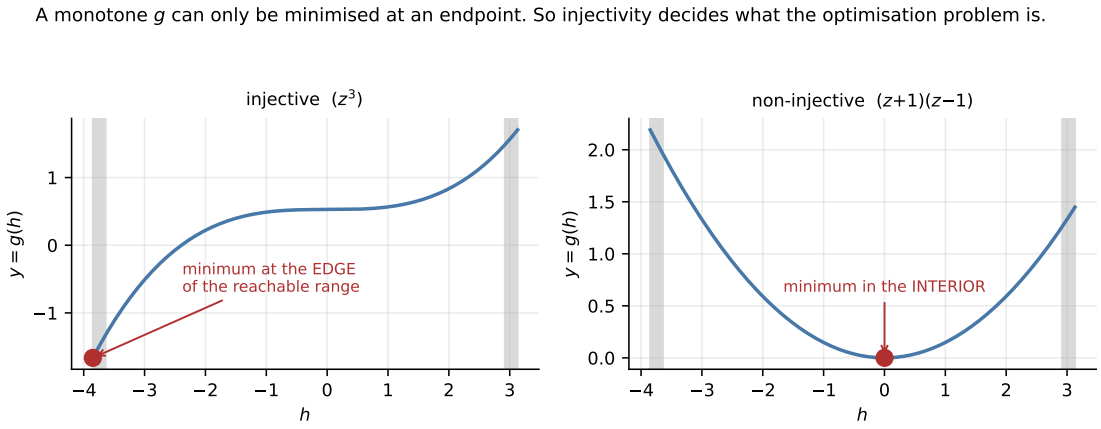


Figure 2: The same point, drawn. Grey bands mark the edges of the range h reaches.

4 What we measured instead

Prediction, with no optimiser anywhere. Both models are fitted to the same n observations and then asked to predict f at 512 held-out points. We score the whole predictive distribution with CRPS, a proper scoring rule, so a model is rewarded for honest uncertainty and not only for a good central guess. Thirty-two fits per map per sample size, at $n = 5, 10, 20, 40$.

5 The result

Two things stand out.

The matched pairs show no detectable effect. This is the comparison the whole design was built for: same degree, same steepness, opposite bijectivity. Note "detectable" — the interval is wide, and Section efsec:confidence quantifies what it could and could not have seen.

Degree	Bijective		Non-bijective		Difference	
3	`mono3`	$+0.0189 \pm 0.0022$	`root3`	$+0.0221 \pm 0.0040$	$+0.003 \pm 0.005$	(0.7 SE)
5	`mono5`	$+0.0367 \pm 0.0098$	`root5`	$+0.1133 \pm 0.0682$	$+0.077 \pm 0.069$	(1.1 SE)

Neither difference is distinguishable from zero.

The linear map is the real special case. `root1` has the smallest gain at every single sample size, four to twenty times below everything else.

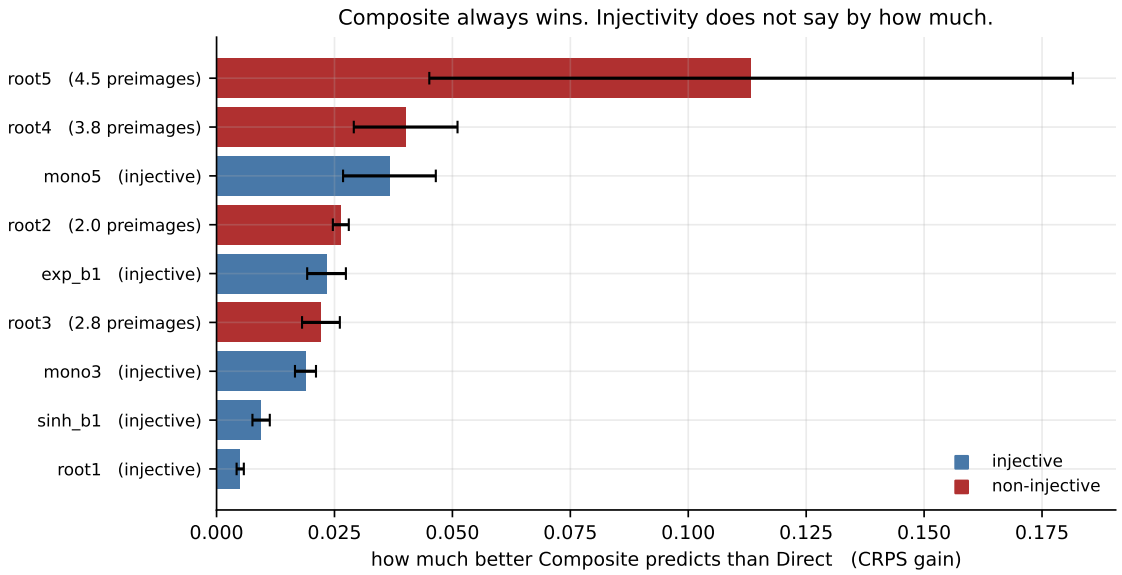


Figure 3: Composite predicts better than Direct everywhere. The blue (bijective) and red (non-bijective) bars are interleaved: extttmono5 and extttexp_b1 are bijective yet outrank three maps that are not. There is a positive across-map association (Section efsec:confidence), but it is not robust and it is confounded with degree.

6 What this means

Bijectivity does control information — that part is proven. If g is bijective, observing h adds exactly nothing; we also confirmed it numerically to six decimal places. That is not in question.

But information is not what the gain is made of. Composite still beats Direct on bijective maps, by a margin several standard errors from zero. The reason is that Direct is fitting the *wrong kind of model*: when g is nonlinear, $f = g(h)$ is not a Gaussian process at all, so a GP placed directly on f is misspecified. Composite's model is correct by construction. Bijectivity removes the information advantage and leaves this one untouched.

So the gain has two sources, and we measured their sum.

Source of the gain	Removed by	Size here
Direct is fitting the wrong model	g being *linear*	the large part
Composite sees information y lacks	g being *bijective*	too small to detect

Linearity kills both, which is why `root1` is lowest, and why Truss and power flow — both linear — show the smallest gains in the paper. Bijectivity kills only the second, which is why the bijective maps still gain.

7 Why a one-dimensional g cannot settle this

The design above pairs each non-bijective map with a bijective one of the same degree, on the reasoning that a Gaussian process should find two degree-5 polynomials about equally hard, so any

difference between them is attributable to bijectivity. That reasoning is wrong, and the way it fails rules out the whole approach.

A monotone g preserves the shape of h; a folding one does not. If g is increasing, $g(h(x))$ has exactly the peaks and valleys that $h(x)$ has, merely relabelled: the objective is a warped copy of the inner function. If g folds — turns around at some interior point — then every time $h(x)$ sweeps past that point the objective turns around too, and $y(x)$ acquires a feature that $h(x)$ never had.

Counting turning points along a line through the input space:

Map		Turns in $y(x)$
h itself	—	1
`root1`, `mono3`, `mono5`, `exp_b1`, `sinh_b1`	all bijective	1
`root3`, `root5`	both non-bijective	2

Every bijective map leaves the shape of h untouched; every non-bijective one adds a fold. The correspondence is exact, and the same holds along the root-spread family: $s = 0$ gives one turn and every $s > 0$ gives two or three.

This is forced, not incidental. For a scalar g,

$$\text{non-injective} \iff \text{non-monotone} \iff g \text{ folds} \iff y(x) \text{ gains turning points},$$

and a stationary GP fits a function with more turning points less well. So in one dimension, **destroying information and making the objective harder to model are the same operation**. They cannot be separated by choosing a better one-dimensional family, because the implication runs through monotonicity and admits no exceptions.

The matched pairs therefore do not do what they were built to do, and neither does the root-spread sweep. Both move injectivity and model difficulty together.

What the sweep can still establish. The confound has a direction, which makes the evidence asymmetric. Spreading the roots raises both the ambiguity and the oscillation, and the oscillation pushes the gain *up*. So:

- a gain that *rises* with $\bar{n}$ is uninterpretable — either cause would produce it;
- a gain that stays *flat* is strong evidence against injectivity mattering, since it holds against a tailwind.

The degree-2 row is also informative, though not in the way first intended: with two roots symmetric about the origin, spreading them shifts the parabola vertically and the affine normalisation removes the shift exactly, so all five maps in that row are the *same problem* to machine precision (2×10^{-16}). Five independent runs of an identical problem measure the Monte Carlo noise floor directly, which calibrates how much movement elsewhere is real.

The escape is more than one intermediate. With $k \geq 2$, g can map two different *vectors* to the same value without folding along any direction. The clearest case is $g(\mathbf{y}) = \sum_i y_i$: the split of the sum is unrecoverable, yet g is a plane, nothing folds, and f remains exactly a Gaussian process so the direct model is correctly specified. Measured, the objective keeps exactly one turning point at every k:

g	$k=2$	4	8	16
$\sum_i y_i$ (linear, correctly specified)	1	1	1	1
$\max_i y_i$ (one kink, no folds)	2	2	3	6
$\sum_i y_i^2$ (folds)	1	0	2	3

Along the top row the number of discarded dimensions grows with k while model difficulty is pinned, which is the separation one dimension cannot provide. Two caveats keep this from being the final answer. For the linear family the discarded information is irrelevant to f — the objective depends only on the sum — so the proposition already predicts exactly zero and the row is a control rather than a discovery, albeit one at the K values of the real benchmarks. For max the discarded information *is* relevant, but its turning-point count still climbs from 2 to 6, so the pinning is partial.

Where this points. Every two-arm design we have tried runs into the same wall from a different direction: regret because bijectivity decides where the optimum can sit, high $\bar{n}$ because no bijective map can be matched on difficulty, and now matched degrees because folding and information loss coincide. The common cause is that Direct differs from Composite in two ways at once, and no choice of g separates them. A model that holds the prior on $\mathbf{h}$ fixed and varies only whether $\mathbf{h}$ is observed differs in one way by construction, at any g — which is the three-arm comparison, arrived at here from a third independent direction.

8 How sure are we of each claim?

What was varied, and how much of it.

Experiment	Random inputs varied	Replicates	Fixed	Metric
Predictive	4 function draws × 8 designs $n \in \{5, 10, 20, 40\}$ training pts	32 fits per map, per n × 4 sizes × 9 maps = 1152	$d = 6$, $\ell = 0.5$ 512 test pts	CRPS, RMSE
Regret	4 function draws × 10 designs (confounded, see §4)	40 BO runs per arm per map × 2 arms × 9 maps = 720	budget 40, $n_0 = 5$ shared designs	final $\log_{10}$ regret
Injectivity $\bar{n}$	—	4000 probe points, 20001-pt grid	draw 0	count of preimages
Exact posteriors	— (deterministic)	no sampling anywhere	$n = 8$–12	Kolmogorov dist.

A "function draw" is a fresh random inner function h; a "design" is a fresh set of training inputs. Varying both matters: with one draw, a conclusion could be a property of that single sampled h and the error bars would never reveal it.

Each claim, with its interval. Intervals are 95%; paired rows use a t interval on the per-fit differences, so the pairing across the shared design is exploited.

Claim	Estimate	95% CI	Confidence
Proved, then confirmed numerically — no sampling involved			
Bijective g: Models 2 and 3 coincide	0.000000	exact	**certain**
Linear g: Models 1 and 3 coincide	0.000000	exact	**certain**
Strong: large t, every replicate agrees			
Composite predicts better than Direct on *every* map	—	—	**strong** (9/9 maps, $t \geq 3.6$)
e.g. `root2`, 32 fits, 32/32 wins	$+0.0263$	$[+0.0229, +0.0298]$	$t = 15.6$
Bijective maps still gain (so the gain is not only information)	$+0.0189$	$[+0.0143, +0.0234]$	$t = 8.5$ (`mono3`)
The linear map gains least, at every n	$+0.0050$	$[+0.0035, +0.0066]$	**strong** (4/4 sizes)
Gain does not shrink as n grows $5 \to 40$	—	—	**strong** (all 9 maps)
Unresolved: the interval is too wide to call			
Degree-3 matched pair: effect of bijectivity	$+0.0033$	$[-0.0057, +0.0123]$	$t = 0.72$ **null**
Degree-5 matched pair: effect of bijectivity	$+0.0767$	$[-0.0584, +0.2118]$	$t = 1.11$ **null**
Across maps, gain correlates with $\bar{n}$	$r = +0.78$	$[+0.25, +0.95]$	**fragile** (see below)
`root5`'s gain	$+0.1133$	$[-0.0258, +0.2525]$	$t = 1.7$ **unreliable**
Confounded by construction — reported, not relied on			
Regret gain, `root1` (linear), 40 runs	$+0.260$	$[-0.107, +0.628]$	n.s.
Regret gain, `root3`, 40 runs	$+0.808$	$[+0.411, +1.205]$	see §4

Why the correlation is called fragile. It rests on one point.

Subset	Pearson r	95% CI	Spearman
All 9 maps	$+0.783$	$[+0.247, +0.952]$	$+0.700$
Dropping `root5` (whose own CI spans zero)	$+0.558$	$[-0.242, +0.906]$	$+0.571$
Polynomials only	$+0.766$	$[+0.030, +0.963]$	$+0.857$

Removing a single map moves the interval across zero. And $\bar{n}$ is itself correlated with degree ($r = +0.48$ across the seven polynomials), so with nine maps the two cannot be separated. Most tellingly: **among the five maps with $\bar{n} = 1.00$ exactly, the gain still spans a factor of** 7.3 (0.0050 to 0.0367). Injectivity held perfectly fixed, something else moves the gain sevenfold.

What the null results could and could not have seen. The degree-3 interval has half-width 0.0090 against a gain of about 0.020, so it would only have detected a bijectivity effect larger than roughly **45% of the gain itself**. The observed 17% difference is well inside the noise. Resolving it needs the standard error down by 2.7×, which is **$\sim$240 fits per map** instead of 32 — about eight times the compute, and still only two matched pairs.

The honest summary. We can say with confidence that the gain is real everywhere, that it is smallest for the linear map, that it survives on bijective maps, and that it does not wash out with data. We *cannot* currently say whether bijectivity has a modest secondary effect: the controlled comparison is null but underpowered, and the across-map correlation is positive but fragile and confounded with degree.

One sentence in the paper should be adjusted. "Even a nonlinear composition adds no information advantage when it is invertible" is correct about *information*, but it reads as a prediction that invertible maps will show no gain. They do show one. On this evidence, how nonlinear g is predicts the gain better than whether it is invertible.

9 What would settle it

Separating the two sources needs a third model: one that holds the correct prior on h but sees only y. Comparing that against Composite isolates the information channel with the misspecification held fixed, which is the only place bijectivity can show up cleanly. That is the three-arm setup, and this experiment is an argument for running it rather than a replacement.

Caveats

- One function family, $d = 6$, single objective, noiseless. The paper's hypervolume benchmarks are none of those things.
- `root5`'s error bar is enormous (± 0.068 on $+0.113$); a few outlier fits dominate it. The degree-3 pair is the trustworthy comparison.
- $\bar{n}$ counts branches without weighting them by how plausible they are, so a map with a technically reachable but very unlikely second branch looks more ambiguous than it behaves.